

AI-Powered Fleet Performance & Delivery Efficiency Dashboard

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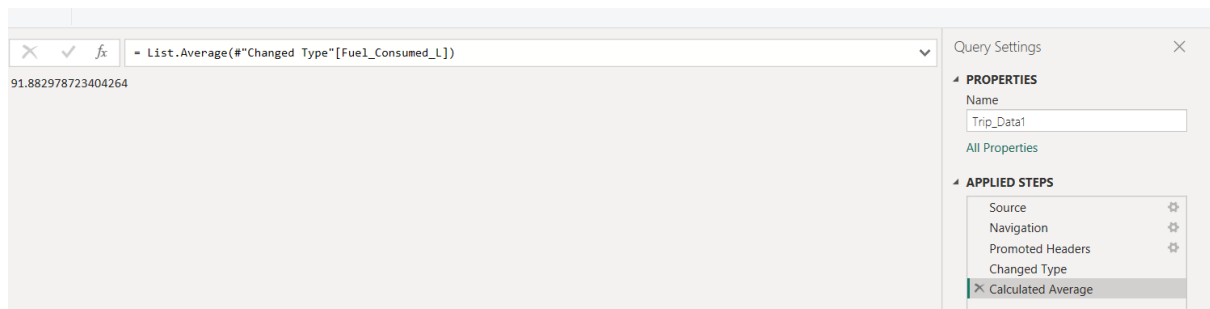
Course: Program In AI Driven Data Analytics

Batch: DA-FNB17

Date: 17-08-2026

1. Data Cleaning & Modelling:

- Fix missing fuel consumption values (use mean imputation). In transformation – statistics-average value is 91.88



But I have used another formula for the Fuel consumption because the Distance is varied form every missing values. We can't fix same average for all missing values.

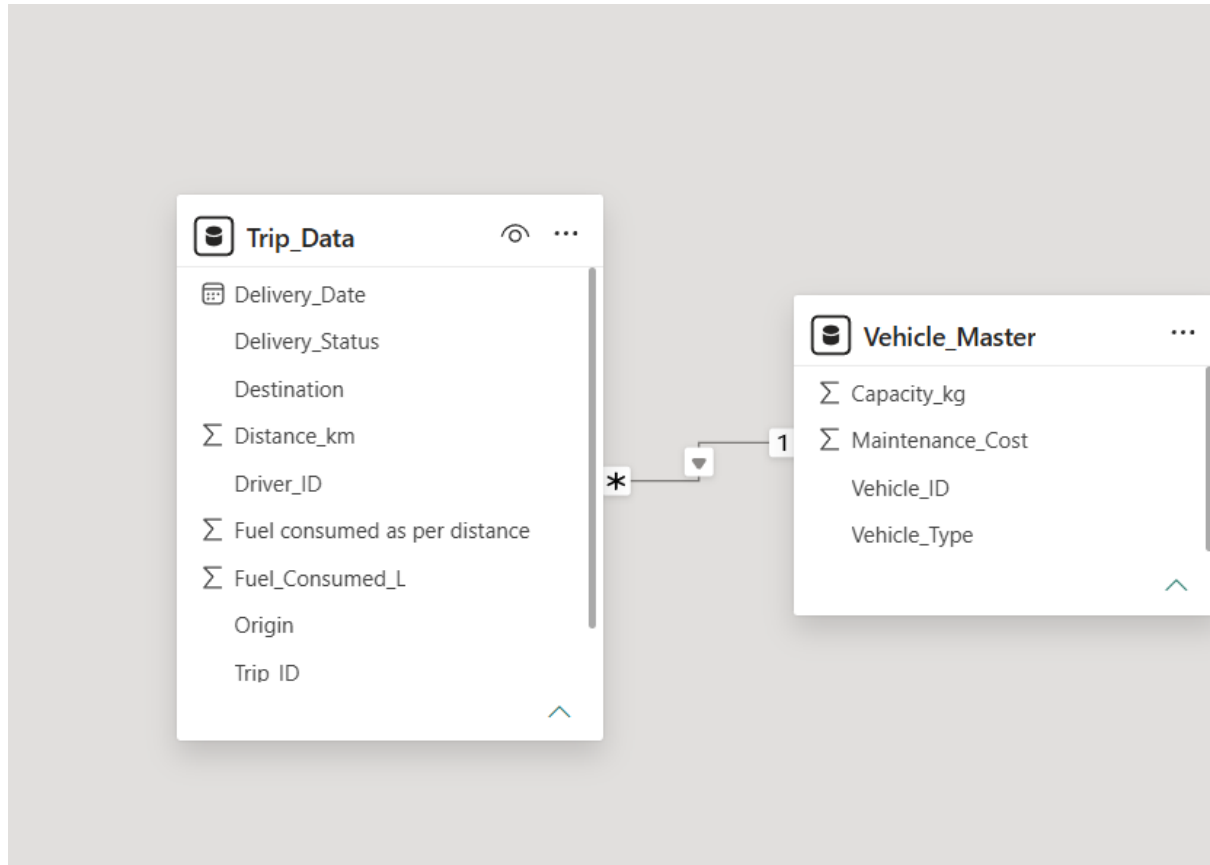
1st we need to find the slope of the fuel and distance `=SLOPE($G$1:$G$51,$F$1:$F$51)`== 0.089397

2nd we need to find the intercept of the fuel and distance `=INTERCEPT($G$2:$G$51,$F$2:$F$51)`== -1.3944

$\text{Fuel consumed_L} = \text{Slope} * \text{Distance_km} - \text{Intercept}$

$\text{Fuel consumed_L} = 0.089397 * \text{Distance_km} - 1.3944$ these values are created in Trip data table.

- Created a relationship using Vehicle_ID between Trip_Data and the Vehicle Master table.



2. DAX Measures:

Fuel Efficiency = Distance / Fuel Consumed, created a new measure named fuel efficiency and divide it by distance and fuel consumed.

re	Formatting	Properties	Sort	Groups
1 Fuel Efficiency = DIVIDE(Trip_Data1[Distance_km],Trip_Data1[Mean imputed Fuel_Consumed_L],blank())				
Vehicle_ID	Driver_ID	Origin	Destination	Distance_km
Mean imputed Fuel_Consumed_L				
Delivery_Status				

On-Time Delivery % = On-Time Trips / Total Trips:

Created a new measure name as Total trips

1 Total Trips = COUNTROWS(Trip_Data1)				
Vehicle_ID	Driver_ID	Origin	Destination	Distance_km

Created a new measure name as on-time trips

1 On-Time Trips = CALCULATE(COUNTROWS(Trip_Data1),Trip_Data1[Delivery_Status]="On-Time")				
Vehicle_ID	Driver_ID	Origin	Destination	Distance_km
Mean imputed Fuel_Consumed_L				
Delivery_Status				
V04	D01	Delhi	Pune	1173
V06	D08	Mumbai	Bangalore	1727
				₹ 108.42
				₹ 161.33
				On-Time
				On-Time

Created a new measure name as On-Time Delivery by dividing the total trips and on time trips

1 On-Time Delivery % = DIVIDE(Trip_Data1[On-Time Trip],Trip_Data1[Total Trips],0)				
Vehicle_ID	Driver_ID	Origin	Destination	Distance_km
Mean imputed Fuel_Consumed_L				

o Cost per km = (fuel cost + Maintenance Cost) / Distance

Note fuel cost =100

First, we bring Maintenance Cost into Trip_Data using the relationship:

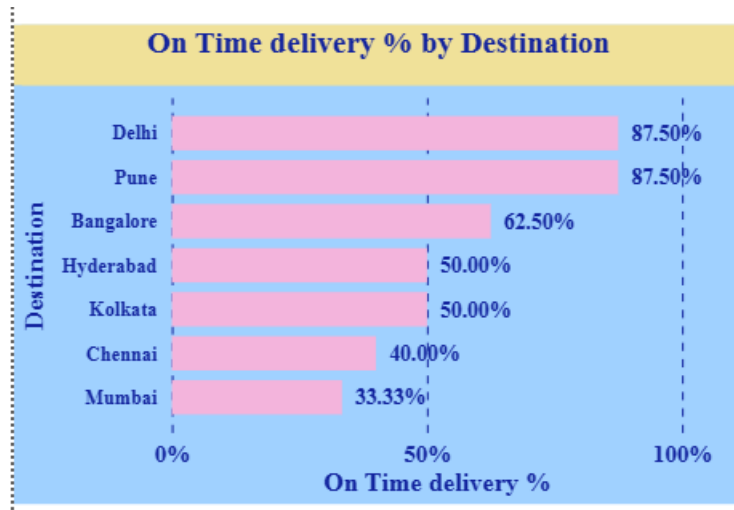
e	Formatting	Properties	Sort
1 Maintenance Cost = RELATED('Vehicle_Master'[Maintenance_Cost])			
Driver_ID	Origin	Destination	Distance_km
Mean imputed Fuel_Consumed_L			
Delivery_Status			

Then we need to create a calculated column Cost per Km in Dax measure

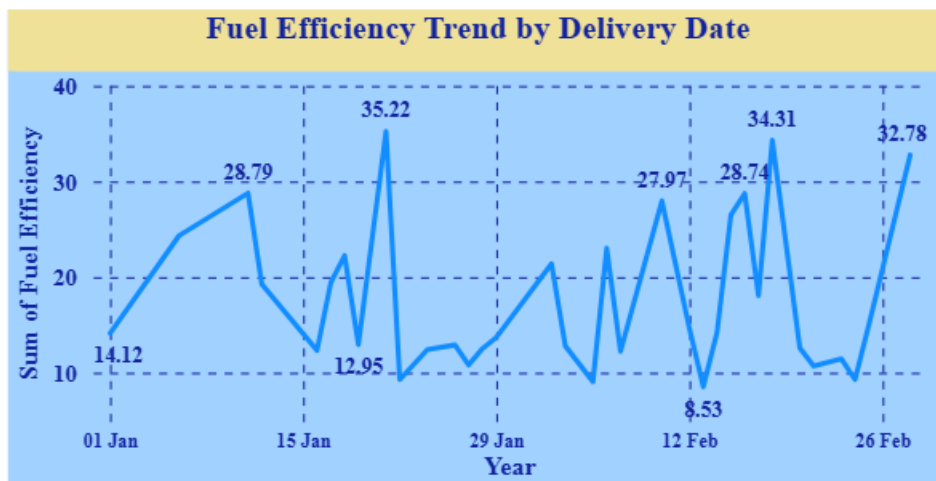
cture	Formatting	Properties	Sort	Groups	F
1 Cost per Km = DIVIDE(100+Trip_Data1[Maintenance Cost],Trip_Data1[Distance_km],0)					
Driver_ID	Origin	Destination	Distance_km	Mean imputed Fuel_Consumed_L	Delivery_Status
					Deliv
D01	Delhi	Pune	1173	₹ 108.42	On-Time
D08	Mumbai	Bangalore	1727	₹ 161.33	On-Time
D08	Mumbai	Pune	1459	₹ 154.7	On-Time
D09	Hyderabad	Pune	382	₹ 26.6	On-Time
					27
					21
					17
					18

3. Visualization:

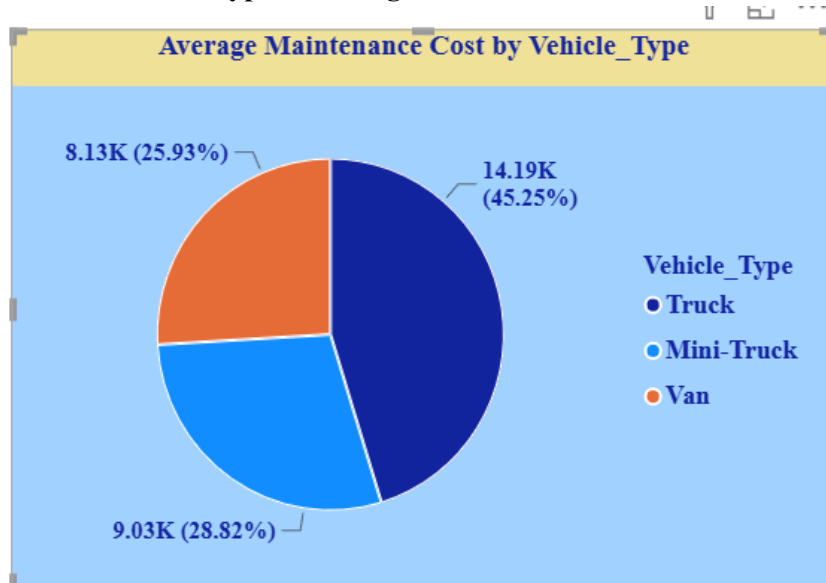
Cluster Bar chart: On-Time Delivery % by Destination.



Line chart: Fuel efficiency trend by delivery date.



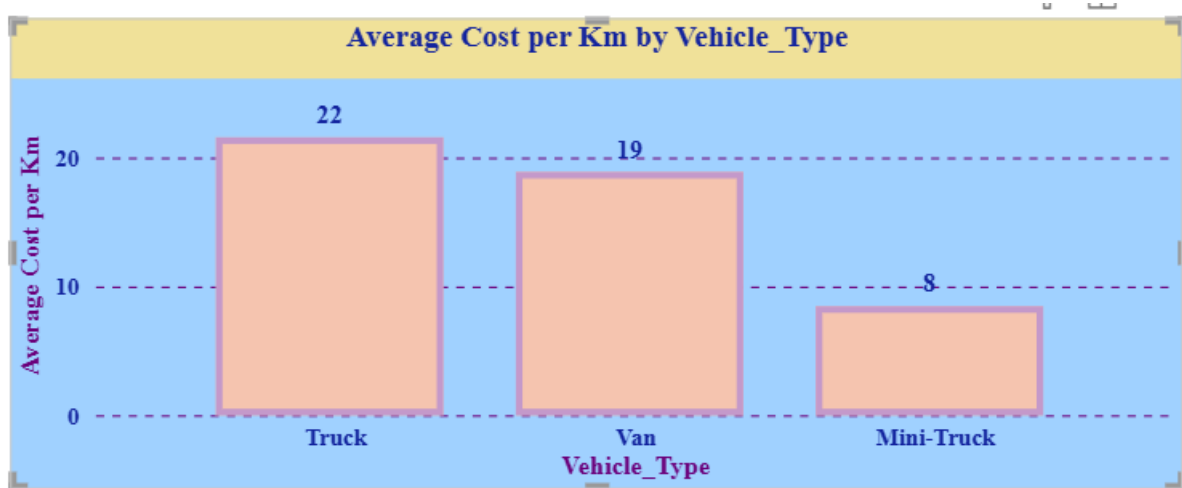
Pie chart: vehicle type vs Average maintenance cost



4. AI-Powered Visuals

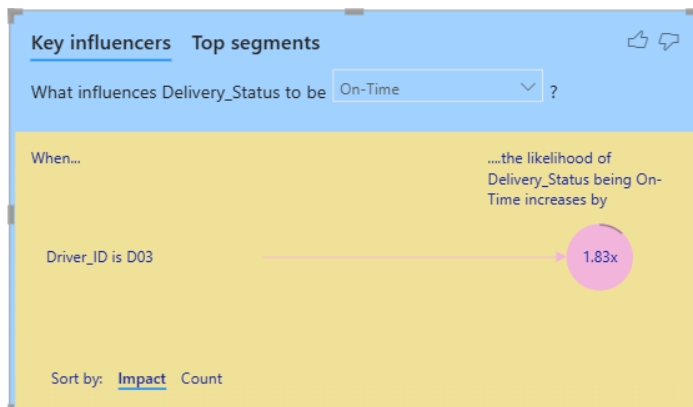
❖ Q&A Visual: ➤ Add Q&A visual → Prompt: “Average Cost per km by vehicle type?”

I don't have Q&A visual in my power bi so I have used the cluster column chart for the average cost per Km by vehicle type



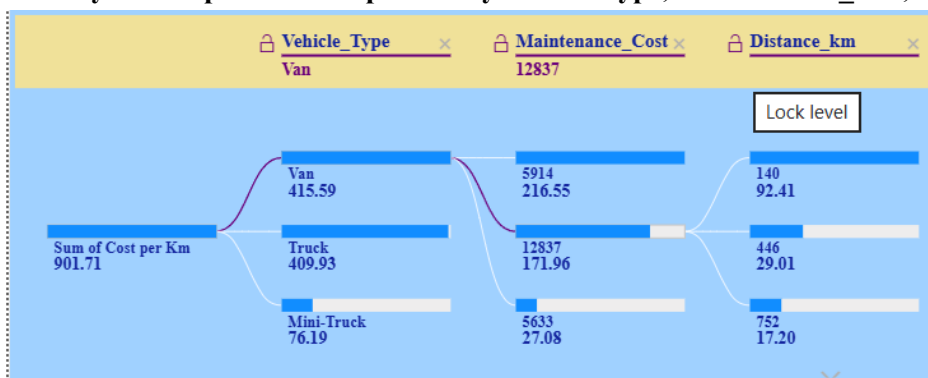
❖ Key Influencers Visual:

➤ Add Delivery Status in the Analyze and explain by -distance in km,vehicle_type, Driver_ID



❖ Decomposition Tree (AI Visual):

➤ Analyze Cost per km → explained by Vehicle Type, Maintenance_cost, and Distance_km.



Fleet Performance & Delivery Efficiency Dashboard

On Time delivery % by Destination

Destination	On Time delivery %
Delhi	87.50%
Pune	87.50%
Bangalore	62.50%
Hyderabad	50.00%
Kolkata	50.00%
Chennai	40.00%
Mumbai	33.33%

Key influencers

What influences Delivery_Status to be

On-Time

When... Driver_ID is D03

Sort by: Impact Count

...the likelihood of Delivery_Status being On-Time increases by 1.83x

Fuel Efficiency Trend by Delivery Date

Delivery Date	Fuel Efficiency
15 Jan	14.12
16 Jan	28.79
17 Jan	35.22
18 Jan	10.5
19 Jan	10.5
20 Jan	10.5
21 Jan	21.5
22 Jan	10.5
23 Jan	27.5
24 Jan	8.53
25 Jan	28.7
26 Jan	34.31
27 Jan	10.5
28 Jan	10.5
29 Jan	10.5
30 Jan	32.78

Average Cost per Km by Vehicle_Type

Vehicle_Type	Average Cost per Km
Truck	22
Van	19
Mini-Truck	8

Average Maintenance Cost by Vehicle_Type

Vehicle_Type	Maintenance Cost	Percentage
Van	18.03	45.94%
Truck	14.19K	35.22%
Mini-Truck	9.03K	22.82%

Cost per Km and Maintenance Cost by Fuel Efficiency

10.01

Goal: 17751 (-99.94%)

